CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

11.2 Fundamental Particles

Syllabus Section: 11. Particle Physics **Paper Type:** Multiple Choice (Paper 1) **Total:** 22 Questions (22 Marks)

Question 1 Easy [1 mark]

What is a particle that is **not** made of any smaller particles?

A elementary particle

B minimum particle

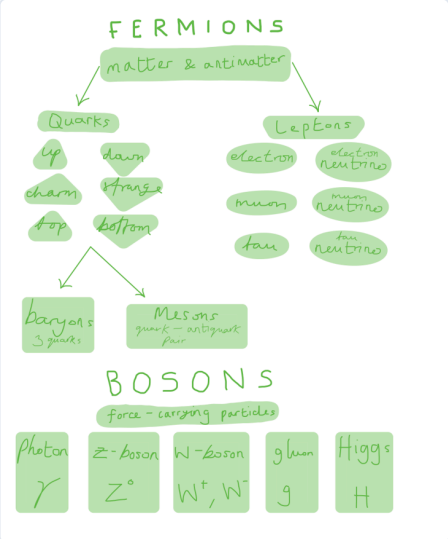
C original particle

D atomic particle

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- An **elementary particle** or **fundamental particle** is a subatomic particle which cannot be broken down further into constituent particles



Question 2 Easy [1 mark]

Light is made up of which type of elementary particle?

A quarks

B leptons

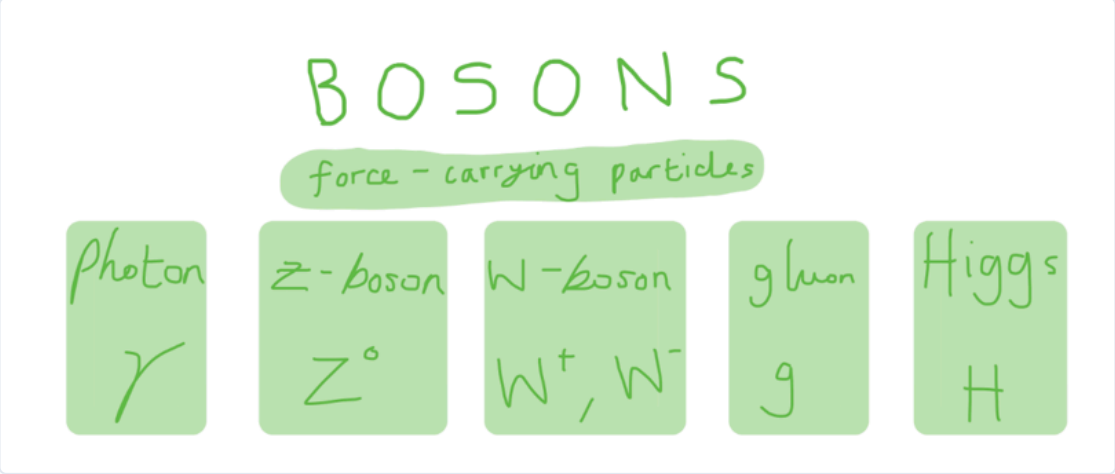
C photons

D bosons

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Photons are the smallest discrete amount or quantum of electromagnetic radiation
  - It is the basic unit of all light
- Photons are always in motion and travel at 3.0×10^8 m s^{-1} in a vacuum
- Photons are a type of boson



A is incorrect because quarks are fermions which cannot exist on their own, instead they combine to make hadrons (e.g. protons, neutrons)

B is incorrect because leptons are fermions (e.g. electrons, neutrinos)

D is incorrect because the question asks for the specific particle that makes up light — photons are the specific particle type. While photons are a type of boson, simply saying bosons is too broad and imprecise as the answer to a question about light

Question 3 Easy [1 mark]

What type of elementary particles are electrons ?

A quarks

B leptons

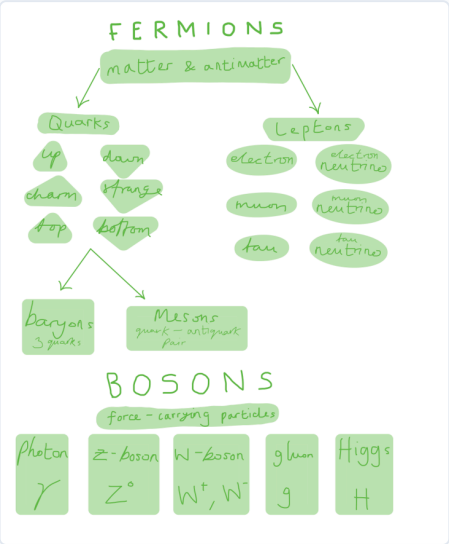
C photons

D bosons

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Electrons are a type of **lepton**



Question 4 Easy [1 mark]

What elementary particles are the building blocks for protons and neutrons ?

A quarks

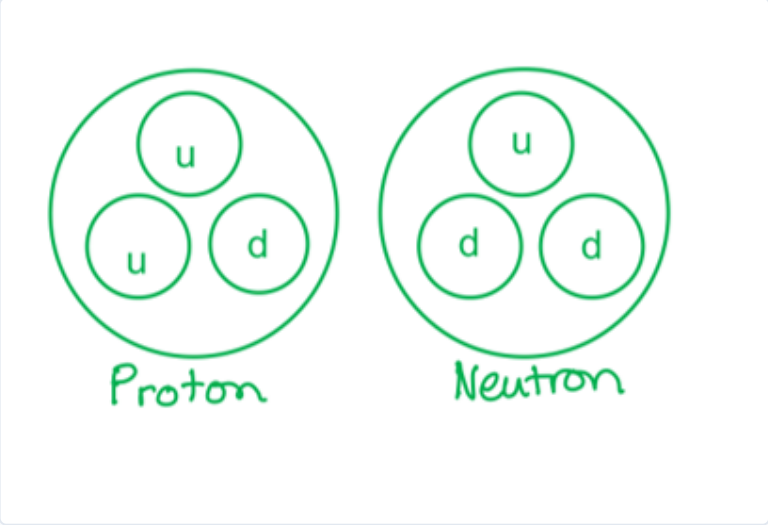
B leptons

C photons

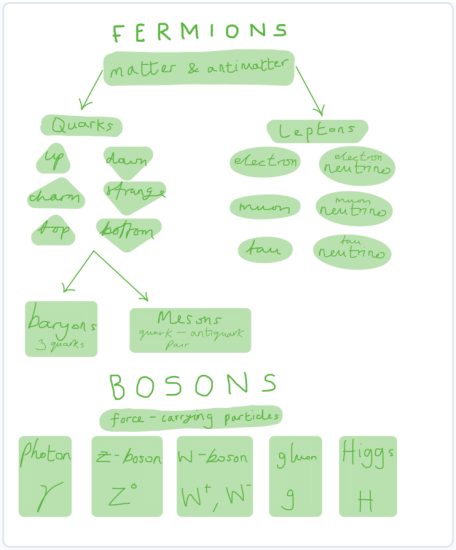
D bosons

Mark Scheme & Explanation:

*The correct answer is **A** because:*



- Quarks are the building blocks for protons and neutrons
- Protons and neutrons are composed of two types of quarks (up and down) held together by gluons :


FERMIONS
matter & antimatter
Quarks
up
down
charm
strange
top
bottom
Leptons
electron
electron neutrino
muon
muon neutrino
tau
tau neutrino
baryons
3 quarks
Mesons
quark – antiquark pair
BOSONS
force-carrying particles
photon
γ
Z-boson
Z°
W-boson
W+, W−
gluon
g
Higgs
H

Question 5 Easy [1 mark]

How many quarks make up a proton ?

A 0

B 1

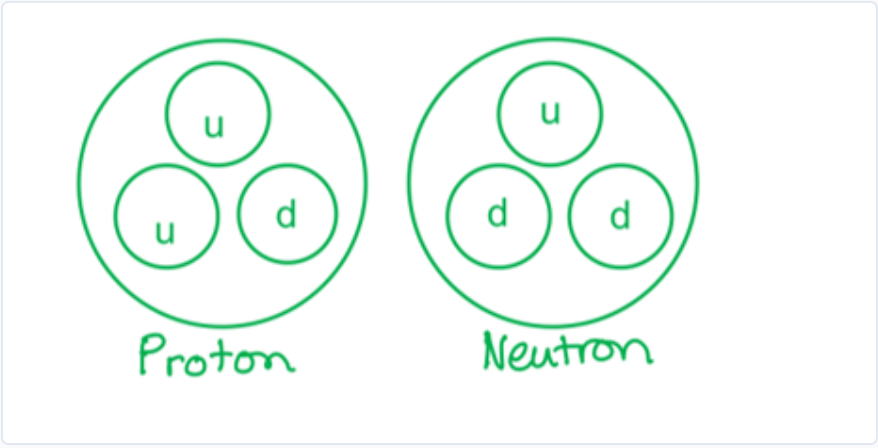
C 2

D 3

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Protons and neutrons are composed of two types of quarks held together by gluons:
  - Up quarks
  - Down quarks
- They are both made of three quarks as shown in the diagram



Question 6 Easy [1 mark]

Which particle is made up of two 'down' quarks and one 'up' quark?

A atom

B electron

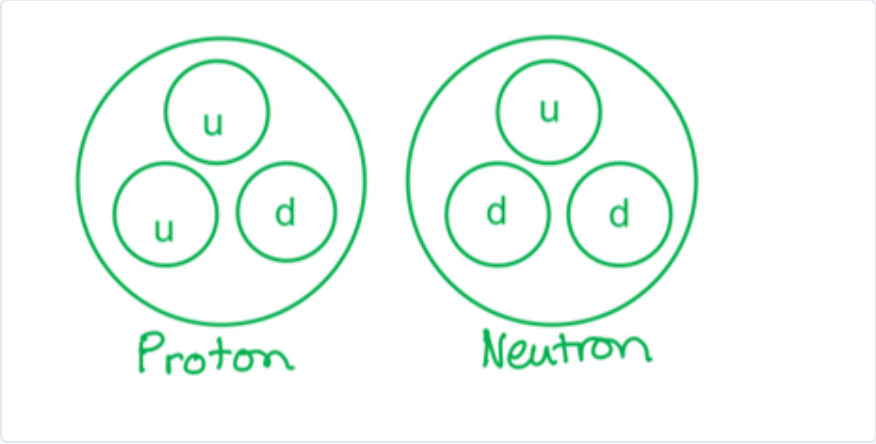
C neutron

D proton

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Protons and neutrons are composed of two types of quarks held together by gluons:
  - Up quarks
  - Down quarks
- They are both made of three quarks as shown in the diagram



- Therefore, neutrons consist of two 'down' quarks and one 'up' quark

Question 7 Easy [1 mark]

How many types of quark are there ?

A 2

B 4

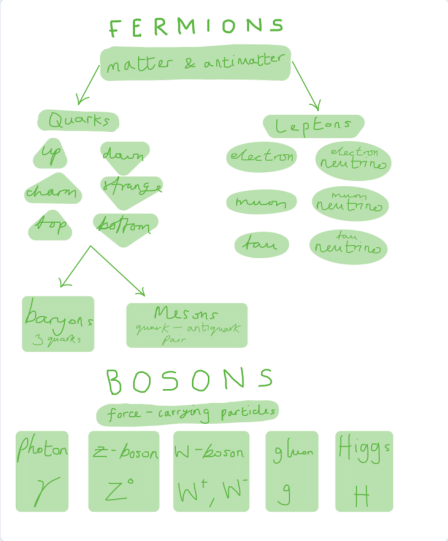
C 6

D 8

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- There are six types, or flavours , of quarks that differ from one another in their mass and charge characteristics
- These **six quark** flavours can be grouped in three pairs:
    - Up and down, charm and strange, and top and bottom



Question 8 Easy [1 mark]

Which elementary particle is a lepton ?

A neutron

B electron

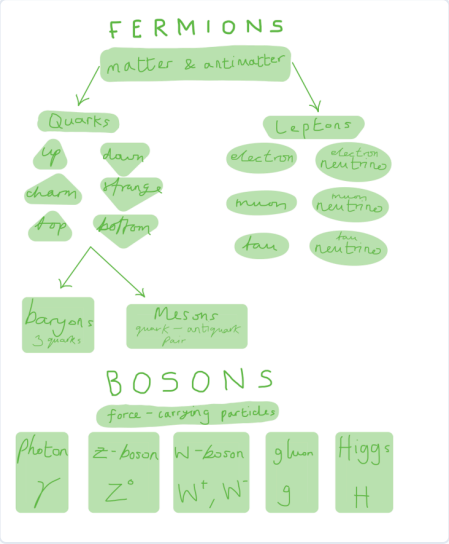
C quark

D proton

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Electrons are a type of **lepton**



Question 9 Easy [1 mark]

What are the fundamental particles of an atom ?

A quarks, gluons and electrons

B protons, neutrons and electrons

C quarks, photons and electrons

D quarks, gluons and neutrino

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Atoms are made up of **elementary particles** or **fundamental particles**
- The elementary particles that make up an atom are quarks, gluons, and electrons
  - Quarks and gluons that make up protons and neutrons and are found in the nucleus of the atom
  - Gluons are a type of boson (force-carrying particle)
  - Electrons are a type of lepton which orbit around the nucleus
- The 12 fundamental particles are as follows :



B is incorrect because protons, neutrons and electrons are not all elementary particles

C is incorrect because photons are not a fundamental particle in an atom

D is incorrect because neutrinos are not a fundamental particle in an atom

Question 10 Easy [1 mark]

What are mesons ?

A a type of composite particle produced by high energy

B a type of gluon

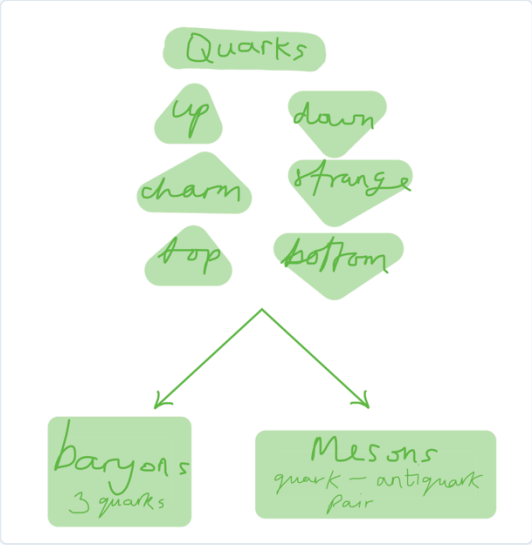
C an antimatter version of the electron

D a type of neutrino

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Hadrons are subatomic particles made from 2 or more quarks which are bound by gluons
- There are two types of hadron:
  - Baryons: three quarks or three antiquarks
  - Mesons: a quark and an anti-quark pair
- They only exist for short periods of time and are produced through very high energy interactions such as those caused by cosmic rays or in particle accelerators
- Therefore, mesons are a type of composite particle produced by high energy



B is incorrect because gluons are a type of boson (force-carrying particle)

C & **D** are incorrect because neutrinos and anti-electrons (positrons) are types of lepton

Question 11 Medium [1 mark]

Which of the following sentences about antimatter is **incorrect** ?

A antimatter is normal matter with an opposite charge

B antimatter is only produced in particle accelerators

C antimatter destroys matter

D equal amounts of antimatter and matter were created during the Big Bang

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Every fermion has an antimatter equivalent e.g. the proton has a quark composition of uud (up, up, down) and the antiproton has a quark composition of $\bar{u}\,\bar{u}\,\bar{d}$ (anti-up, anti-up, anti-down)
    - Antimatter is produced synthetically in particles accelerators for experiments
    - It is also produced naturally through the radioactive decay of potassium 40, cosmic ray interactions in the atmosphere and thunderstorms
    - It has also been detected in the Earth's Van Allen radiation belts
- Therefore, antimatter does not just exist as a result of particle accelerator experiments

A is incorrect because antimatter is normal matter but with an opposite charge, for example, the antimatter particle of an electron, which has a negative charge, is the positron, which has a positive charge - both particles are identical in every other way

C is incorrect because matter and antimatter annihilate each other when they come into contact

D is incorrect because equal amounts of matter and antimatter would have been created in the Big Bang (however, it is still unknown why we live in a universe dominated by matter with almost zero antimatter)

Question 12 Medium [1 mark]

Fundamental or elementary particles are particles that aren't made up of smaller particles.

What is the most common type of fundamental particle in the universe ?

A atom

B meson

C neutrino

D quark

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Trillions of neutrinos are produced every second in nuclear reactions at the center of stars and in supernova explosions when stars die

A is incorrect because atoms are not fundamental particles as they are made up of quarks and electrons

B is incorrect because mesons are not fundamental particles as they are made up of quarks

D is incorrect because while a quark is a fundamental particle, it is not found in as high numbers as the neutrino

Question 13 Medium [1 mark]

In a nucleus of Tritium, ${}^{3}_{1}H$, how many 'down' quarks are there?

A 5

B 4

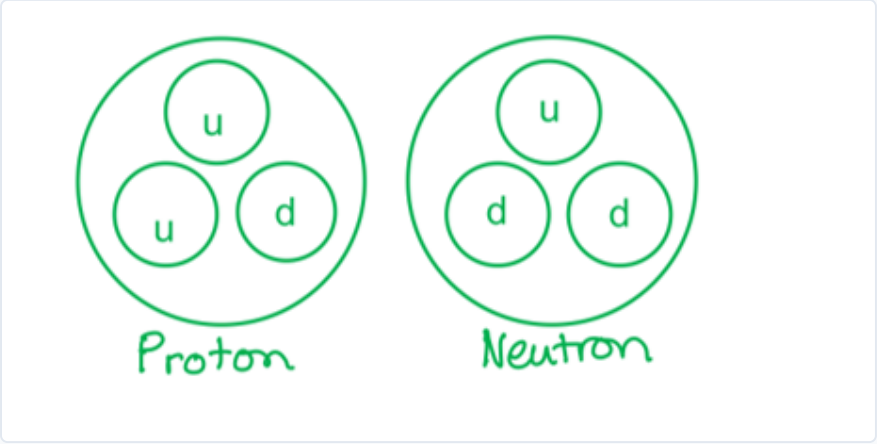
C 3

D 2

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Protons and neutrons are composed of two types of quarks held together by gluons:
  - Up quarks
  - Down quarks
- They are both made of three quarks as shown in the diagram



- Tritium is known as hydrogen-3, therefore, it has 2 neutrons and 1 proton
  - Each neutron has two down quarks, making a total of 4
  - The proton has one down quark, giving an overall total of 5
- Therefore, a tritium nucleus contains 5 'down' quarks

Question 14 Medium [1 mark]

Which row in the table gives the correct quark combination for the particle?

	Particle	Category	Quark combination
A	neutron	baryon	ud
B	neutron	meson	udd
C	proton	baryon	uud
D	positive pion	meson	ud

A

B

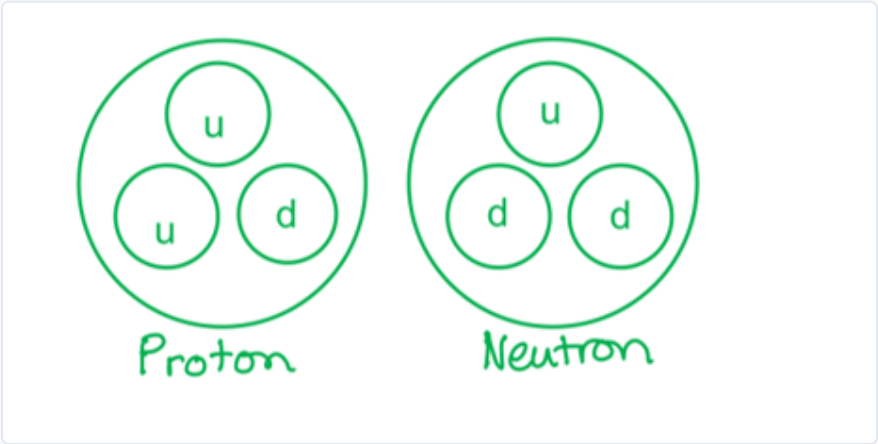
C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- There are two types of hadron:
  - Baryons: three quarks or three antiquarks
  - Mesons: a quark and an anti-quark pair
- Protons and neutrons are baryons composed of up and down quarks held together by gluons:
- They are both made of three quarks as shown in the diagram



- Therefore, option **C** is correct

A is incorrect because a neutron has three quarks, two down and one up

B is incorrect because a neutron is not a type of meson; a meson contains one quark and one anti-quark bound by a gluon

D is incorrect because a positive pion is a meson but consists of one u and one d quark

Question 15 Medium [1 mark]

What are the numbers of hadrons, baryons and mesons in an atom of $^{7}_{3}Li$?

	Hadrons	Baryons	Mesons
A	7	3	3
B	7	4	4
C	7	7	0
D	10	7	0

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because :*

- Hadrons are quark-based particles, and protons and neutrons are both made of different types of quarks
  - Baryons are a type of hadron
- Lithium contains three protons and four neutrons giving a total of seven hadrons and baryons

A & **B** are incorrect because a meson contains one quark and one anti-quark bound by a gluon; lithium does not contain any mesons

Question 16 Hard [1 mark]

Quarks are thought to make up protons and neutrons.

The 'up' quark has a charge of $\frac{2}{3}e$; a 'down' quark has a charge of $-\frac{1}{3}e$, where *e* is the elementary charge (+ 1.6 × 10^{-19} C)

How many up quarks and down quarks must a proton contain ?

	Up quarks	Down quarks
A	0	3
B	2	1
C	1	2
D	1	1

A

B

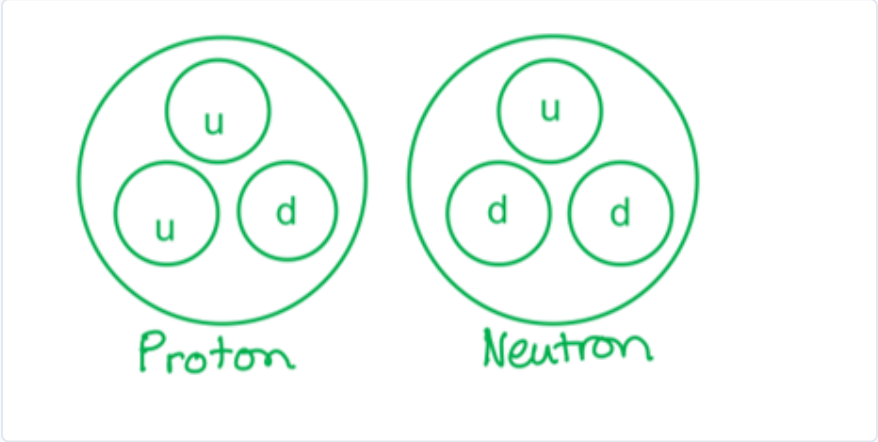
C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Protons and neutrons are baryons composed of up and down quarks held together by gluons
- They are both made of three quarks as shown in the diagram



- A proton consists of two up quarks and one down quark,
  - $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +e$
- This gives an overall charge of +*e*, therefore, a proton has a quark combination of uud

Question 17 Hard [1 mark]

A CERN detector shows a signal only in the hadronic calorimeter. No signal is observed in the tracker, electromagnetic calorimeter or muon chambers.

Which particle must be responsible for this signal?

A pion

B neutrino

C photon

D neutron

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The hadronic calorimeter measures the energy of hadrons
  - Hadrons are particles made of quarks such as baryons or mesons
- No electromagnetic signal was detected, so, the particle must not have a charge
- The only hadron in the list is a neutron; a neutron is made up of two down quarks and one up quark and is neutrally charged overall
  - Therefore, the detected particle must be a neutron

A is incorrect because a pion is a type of meson; a meson contains one quark and one anti-quark bound by a gluon

B is incorrect because neutrinos are leptons with no charge

C is incorrect because photons are the smallest discrete amount or quantum of electromagnetic radiation; it is the basic unit of all light

Question 18 Hard [1 mark]

Some particles are made up of a combination of three quarks.

Which particle would **not** give a particle a charge of either +1.6 × 10^{-19} C or zero?

A up, up, up

B up, up, down

C up, strange, strange

D up, down, down

Mark Scheme & Explanation:

*The correct answer is **A** because:*

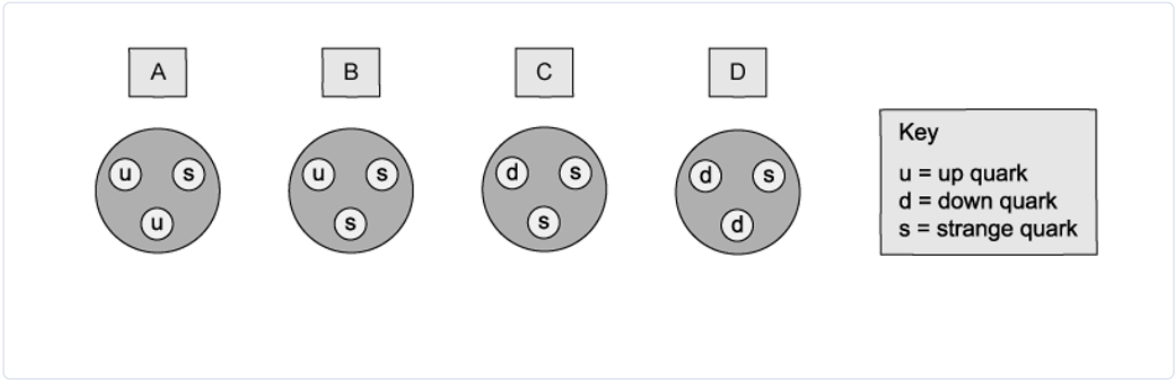
- The 'up' quark has a charge of $\frac{2}{3}e$
- The 'down' quark has a charge of $-\frac{1}{3}e$
  - Where *e* is the elementary charge, + 1.6 × 10^{-19} C
- The 'strange' quark has a charge of $-\frac{1}{3}e$
- Consider option **A**:
  - $\frac{2}{3}e + \frac{2}{3}e + \frac{2}{3}e = +2e$
- This is correct as it not neutral as it is double + 1.6 × 10^{-19} C

B is incorrect because	$\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +e$ a quark combination of up, up down is the combination in a proton and would have a charge of + 1.6 × 10^{-19} C
C is incorrect because	$\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e = 0$ a quark combination of up, strange, strange would give a charge of zero
D is incorrect because	$\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e = 0$ a quark combination of up, down, down would give a charge of zero

Question 19 Hard [1 mark]

The composition of four different hadrons are shown in the diagram below. One of the hadrons is a Σ^+ particle. It has a charge of $+e$, where is e is the elementary charge.

Which hadron could be the Σ^+ particle ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Hadrons are particles are particles made of quarks such as baryons or mesons
- The 'up' quark has a charge of $\frac{2}{3}e$
- The 'down' quark has a charge of $-\frac{1}{3}e$
- The 'strange' quark has a charge of $-\frac{1}{3}e$
  - Where e is the elementary charge, $+ 1.6 \times 10^{-19}$ C
- Consider option **A**:
  - $\frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = +e$
- Therefore, the only option that would have a charge of $+ 1.6 \times 10^{-19}$ C is **A**

B is incorrect because	$\frac{2}{3}e - \frac{1}{3}e - \frac{1}{3}e = 0$ a quark combination of up, strange, strange would give a charge of zero
C is incorrect because	$-\frac{1}{3}e - \frac{1}{3}e - \frac{1}{3}e = -1e$

	a quark combination of down, strange, strange would be a Σ^- baryon with a −1 charge
D is incorrect because	$-\frac{1}{3}e - \frac{1}{3}e - \frac{1}{3}e = -1e$ a quark combination of down, down, strange would be a Σ^- particle with a charge of −1

Question 20 Hard [1 mark]

When high-speed hadrons collide in particle accelerators, a positive pion (π^+) is an unstable particle produced. The π^+ particle has a charge of $+e$.

What is the quark combination of the π^+ particle?

A $u \quad u$

B $u \quad d$

C $d \quad u$

D $d \quad d$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- A meson contains one quark and one anti-quark bound by a gluon
- The 'up' quark has a charge of $\frac{2}{3}e$
  - Where e is the elementary charge + 1.6×10^{-19} C
- The anti-'down' quark has a charge of $\frac{1}{3}e$
  - $\frac{2}{3}e + \frac{1}{3}e = +1e$
- Therefore, the π^+ particle must have one up quark and one anti-down quark to have an overall charge of $+1e$

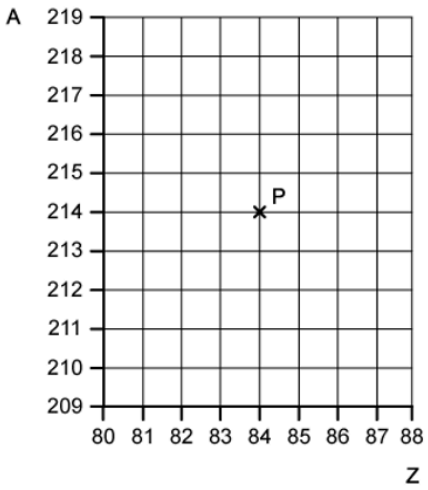
A is incorrect because a meson with a $u \quad u$ combination would be a π^0 particle

C is incorrect because a meson with a $d \quad u$ combination would be a π^- particle

D is incorrect because a meson with a $d \quad d$ combination would be a π^0 particle

Question 21 Hard [1 mark]

A graph of nucleon number A against proton number Z is shown in the diagram below.



The graph shows a cross that represents a nucleus P.

Nucleus P decays by emitting an α-particle to form a nucleus Q.

Nucleus Q then decays by emitting a β^- particle to form a nucleus R.

The quark composition of one nucleon in Q is changed during the emission of the β^- particle.

What is the corresponding change in quark composition from nucleus Q to nucleus R ?

A no change

B *u*-quark to *d*-quark

C *d*-quark to *u*-quark

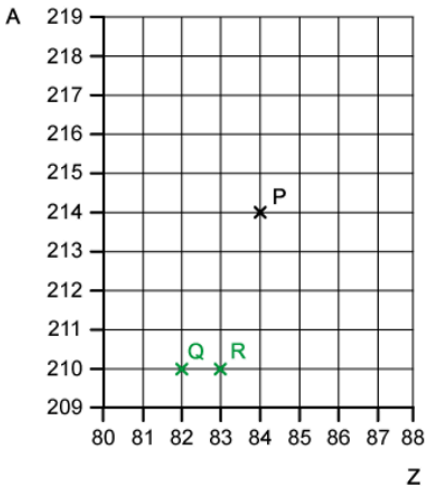
D *d*-quark to *s*-quark

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Nucleus P has a nucleon number A = 214 and a proton number Z = 84
- It decays via α-emission to form nucleus Q:
  - $^{214}_{84}P \rightarrow {}^{210}_{82}Q + {}^{4}_{2}\alpha$

- Overall, the emission of an α-particle from a nucleus will cause the proton number to decrease by two and the nucleon number to decrease by four, as shown on the diagram
    - So, nucleus Q has a nucleon number A = 210 and a proton number Z = 82
- It then decays via β-emission to form nucleus R
    - $^{210}_{82}Q \rightarrow {}^{210}_{83}R + {}^{0}_{-1}\beta$
- Overall, the emission of a β-particle from a nucleus will cause the proton number to increase by one and the nucleon number to stay the same, as shown on the diagram
    - So, nucleus R has a nucleon number A = 210 and a proton number Z = 83
- During the decay of Q into R, a neutron is changed into a proton
    - Quark composition of neutron = udd
    - Quark composition of proton = uud
- Therefore, a d-quark has changed into a u-quark



Question 22 Hard [1 mark]

The equation for β^- decay is

$$n \rightarrow p + e^- + v_e$$

Using your knowledge of antiparticles and the weak interaction, what is the equation for β^+ decay?

A $n \rightarrow p + e^+ + v_e$

B $p \rightarrow n + e^+ + v_e$

C $n + e^+ \rightarrow p + v_e$

D $p \rightarrow n + e^+ + v_e$

Mark Scheme & Explanation:

*The correct answer is **B** because*

- Protons and neutrons are not antiparticles of each other, instead the weak interaction between quarks gives rise to a change in quark composition
  - Quark composition of neutron $= udd$
  - Quark composition of proton $= uud$
- In β^- decay, a neutron turns into a proton $udd \rightarrow uud$
- Whereas, in β^+ decay, a proton turns into a neutron $uud \rightarrow udd$
- Since electrons and neutrinos are both leptons, they are fundamental particles
- The a β^- particle is simply an electron (e^-), and a β^+ particle is its antiparticle – a positron (e^+)
- In β^- decay, an anti-electron neutrino is produced, so, in β^+ decay an electron neutrino is produced
- Therefore, the overall equation for β^+ decay is:
  - $p \rightarrow n + e^+ + v_e$

A is incorrect because in beta-minus and beta-plus decay, a neutron turns into a proton or a proton turns into a neutron due to the weak interaction, their antiparticles are not involved

C is incorrect because as it is a decay equation, the β^+ particle must be on the right hand side as a product of the reaction

D is incorrect because in β^- decay, an anti-electron neutrino is produced, so, in β^+ decay an electron neutrino is produced, an antiparticle is represented by the line above the symbol